

29.6 Summary and Notes

29.6.1 Summary

Idea.

1. The training error of any fitting procedure understates its in-sample prediction error by $2n^{-1} \sum_i \text{Cov}(\hat{\mu}_i, Y_i)$ on average. For least squares on p columns this is $2\sigma^2 p/n$, whether or not the model is correct, and for a linear smoother $2\sigma^2 \text{tr}(\mathbf{S})/n$ (Theorem 29.1.1).
2. The degrees of freedom of a procedure are $\sigma^{-2} \sum_i \text{Cov}(\hat{\mu}_i, Y_i)$ (Definition 29.1.2). Under normality they equal the expected divergence of the fit, by Stein's lemma (Lemma 27.5.1, Proposition 29.1.2). Soft thresholding costs its expected number of nonzero coefficients, hard thresholding and subset search cost more (Proposition 29.1.3 and Example (Best subset of a given size)).
3. Mallows' C_p is an unbiased estimate of the scaled risk of each candidate, and it prefers a larger model iff the added columns have $F > 2$ (Definition 29.2.1 and Proposition 29.2.1). AIC estimates a Kullback–Leibler discrepancy, and AICc estimates it without bias in the normal linear model (Definition 29.2.2 and Theorem 29.2.2). BIC approximates a log marginal likelihood, with penalty $\log n$ per parameter.
4. Criteria with a growing penalty such as BIC find a true finite model with probability tending to one. AIC keeps each irrelevant column with probability about $\Pr\{\chi^2(1) > 2\}$ however large n is, but it is efficient for prediction when no candidate is true (Proposition 29.2.4).
5. In an orthonormal design selection is hard thresholding, and no threshold dominates another (Proposition 29.2.5 and Eq. 29.2.4).
6. K -fold cross-validation estimates the expected error of the procedure fitted to $n - n/K$ cases, not the error of the model at hand (Proposition 29.3.1). For quadratic penalties, leave-one-out needs no refitting, and GCV replaces the leverages by their average (Theorem 29.3.2 and Definition 29.3.2). Every data-driven step must be repeated inside the folds (Example (Screening, then cross-validating)).
7. Criteria that depend on SSE agree within a size (Proposition 29.4.1). Branch and bound finds the best subset exactly with far fewer fits (Proposition 29.4.2). Stepwise steps maximize partial correlations (Proposition 29.4.3) and can miss the best subset, and the chosen subset is unstable (Example (Bootstrapping the state regressions)).
8. After selection, estimates are biased away from zero, minimized error estimates are optimistic, and naive intervals under-cover (Proposition 29.5.1). Sample splitting (Proposition 29.5.2), simultaneous intervals over all submodels (Proposition 29.5.3) and conditional intervals (Example (A selective interval)) restore validity at different prices.

29.6.2 Notes and sources

Coverage. The chapter covers Seber and Lee (2003, chapter 12), Christensen (2020, chapter 14), Sen and Srivastava (1990, chapter 11) and Fahrmeir, Kneib, Lang and Marx (2021, section 3.4) with its own arguments and examples. The comparison of short and long regressions behind selection is Theorem 19.1.5. Shrinkage, the high-dimensional case, and penalized and boosted fits are in [Chapter 27](#), [Chapter 28](#) and [Chapter 30](#); bagging and random forests are outside the book's scope.

Optimism and degrees of freedom. The covariance form of the optimism is due to Efron (1986, 2004), and the covariance definition of degrees of freedom, with the observation that selection adds to them, to Ye (1998). Stein (1981) proved the identity behind Proposition 29.1.2 and the unbiased risk estimate. Zou, Hastie and Tibshirani (2007) showed that the number of nonzero lasso coefficients is unbiased for its degrees of freedom, and Tibshirani (2015) studied the search degrees of freedom of best subset selection, of which Proposition 29.1.3 is the orthonormal case. Breiman (1992) estimated the prediction error of a selected model by resampling residuals, and Breiman and Spector (1992) compared fixed and random designs and leave-one-out with five- and ten-fold cross-validation.

Criteria. C_p and its plot are due to Mallows (1973), AIC to Akaike (1973, 1974), and AICc to Sugiura (1978) and Hurvich and Tsai (1989), whose derivation Theorem 29.2.2 follows. BIC is from Schwarz (1978); Kass and Raftery (1995) and Kass and Wasserman (1995) discuss its accuracy. Consistency for penalties with $c_n \rightarrow \infty$, $c_n/n \rightarrow 0$ is due to Nishii (1984), efficiency to Shibata (1981) and Li (1987), and their incompatibility to Yang (2005). Konishi and Kitagawa (2008) treat information criteria at book length. The pretest estimator goes back to Bancroft (1944).

Cross-validation. Cross-validation is due to Stone (1974) and Geisser (1975), PRESS to Allen (1974), the equivalence with AIC to Stone (1977), and the need for large validation sets for consistency to Shao (1993); Shao (1997) unifies these results. GCV is from Craven and Wahba (1979) and Golub, Heath and Wahba (1979), and the one-standard-error rule from Breiman, Friedman, Olshen and Stone (1984). Bengio and Grandvalet (2004) and Bates, Hastie and Tibshirani (2024) analyse its variance and its target. Varma and Simon (2006) and Hastie, Tibshirani and Friedman (2009, section 7.10) discuss cross-validation after screening.

Search. Enumeration by sweeps is due to Schatzoff, Tsao and Fienberg (1968), branch and bound to Furnival and Wilson (1974), and stepwise regression to Efroymson (1960). Miller (2002) is the standard monograph on subset selection. Natarajan (1995) proved NP-hardness, Bertsimas, King and Mazumder (2016) used mixed-integer optimization, and Hastie, Tibshirani and Tibshirani (2020) compare best subset, forward stepwise and the lasso. Breiman (1996b) analysed instability, Breiman (1996a) proposed bagging, Hoeting, Madigan, Raftery and Volinsky (1999) review model averaging, and Meinshausen and Bühlmann (2010) introduced stability selection.

Inference after selection. The screening paradox is Freedman's (1983). Leeb and Pötscher (2005, 2006) showed that post-selection distributions cannot be estimated uniformly. Sample splitting goes back to Cox (1975), with modern versions by Wasserman and Roeder (2009) and Rinaldo, Wasserman and G'Sell (2019). Proposition 29.5.3 is the PoSI approach of Berk, Brown, Buja, Zhang and Zhao (2013). Exact conditional inference is due to Lee, Sun, Sun and Taylor (2016) and Tibshirani, Taylor, Lockhart and Tibshirani (2016), and data carving to Fithian, Sun and Taylor (2014).

29.6.3 References

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